Storm Preparedness AI: Data Strategy & Data Sources

Modeling framework, source inventory, and immediate team decisions

Data Strategy

Modeling period: 2014–2022, primarily because our downloaded EAGLE-I outage data covers this period.

Basic Combination

- **HURDAT2** → storm track, wind, intensity
- **ONI** → El Niño/ENSO condition
- **FEMA NRI** → county hazard/exposure characteristics
- **CDC SVI** → county social vulnerability
- **EAGLE-I + MCC** → actual outage % (target)

Primary join logic: county FIPS + storm/date/time.

- **Historical NHC forecasts** → used separately for realistic pre-landfall replay/testing.
- **FEMA shelters** → used later for resource recommendations, not outage-model training.

Team Action Items

1. Finalize geography: Atlantic/Gulf only or include Eastern Pacific?
2. Finalize years: Start with 2014–2022 or obtain newer EAGLE-I data?
3. Define target: e.g., peak % customers without power per county.
4. Finalize model features: Decide which NRI/SVI variables to use.
5. Test ONI value: Compare the same outage model with vs. without ONI.
6. Lock MVP: Outage prediction → county risk → map → basic resource recommendations.

Source 1: NHC HURDAT2

HURDAT2 is a post-storm best-track dataset. It represents NHC's later assessment of what happened. It is not the forecast that was available before landfall.

Source: NOAA National Hurricane Center (NHC)

Main page: NHC Data Archive

A. Atlantic HURDAT2

Dataset: Atlantic Hurricane Database (HURDAT2), 1851–2025

Coverage: Atlantic Ocean, Caribbean, and Gulf

Frequency: Primarily 6-hour intervals, with some additional observations such as landfall or peak intensity.

Format Documentation: Atlantic HURDAT2 Format Documentation

B. Northeast/North Central Pacific HURDAT2

Dataset: Northeast and North Central Pacific Hurricane Database (HURDAT2),1945-2025

Coverage: Northeast and North Central Pacific

Frequency: Primarily 6-hour intervals.

Format Documentation: NE/NC Pacific HURDAT2 Format Documentation

Contains

- Storm ID and name
- Date and time
- Storm status/type
- Latitude and longitude
- Maximum sustained wind
- Minimum central pressure
- Wind radii / storm size for periods where available
- Special observations such as landfall

Source 2: Historical NHC Forecast/Advisory Data

Source: NOAA National Hurricane Center (NHC)

Dataset: Historical Tropical Cyclone Forecasts / Advisories (specific advisories still need to be extracted)

Coverage: Historical hurricane seasons, organized by year and storm

Contains

- Storm-specific advisories issued during the event
- Forecast storm position and track
- Forecast intensity
- Forecast times
- Watches and warnings
- Forecast graphics/cones
- Other storm-specific advisory products

Source 3: EAGLE-I Power Outage Data

This dataset provides the historical local impact/target variable for the power-outage prediction model.

A. ORNL EAGLE-I Historical Outage Data

Source: Oak Ridge National Laboratory (ORNL), EAGLE-I

Dataset: EAGLE-I Power Outage Data 2014–2022

Frequency: 15-minute intervals

Coverage: 2014–2022 in the downloaded package.

Contains

- **fips_code** Unique County FIPS code
- **county** county
- **state** State name
- **sum** Number of electric customers without power
- **run_start_time** Timestamp when outage was recorded
- **Timestamps** All timestamps are reported in UTC.

Supplementary file: coverage_history.csv

- Describes how much of each state's electric-customer population was covered by EAGLE-I during different years.

B. Figshare EAGLE-I Supplementary Data

Source: Figshare repository associated with the published EAGLE-I dataset

This has one additional supplementary file: MCC.csv.

MCC.csv (Modeled County Customers): Contains the modeled number of electric customers in each county as of 2022, with 3,235 county records.

Source 4: FEMA National Risk Index (NRI)

FEMA's NRI provides county-level and census tract information related to risk from 18 natural hazards, including hurricanes.

Source: Federal Emergency Management Agency (FEMA)

Dataset: National Risk Index (NRI)

Geographic level: U.S. County

Contains

- **Geographic identifiers:** State, state abbreviation, state FIPS, NRI ID
- **County characteristics:** Population, building value, agricultural value, land area
- **Overall risk measures:** Expected Annual Loss (EAL) score, rating, and loss estimates
- Hazard-specific measures for 18 natural hazards, including hurricanes
- For each hazard, the dataset generally provides:
  - Historical event frequency/count
  - Area exposed

- Population exposed
- Building/property exposure
- Agricultural exposure, where applicable
- Expected Annual Loss
- Annualized Loss Rate
- Risk scores/percentiles

Source 5: NOAA Oceanic Niño Index (ONI)

Source: NOAA Climate Prediction Center (CPC)

Dataset: NOAA CPC Oceanic Niño Index

Contains

- Year
- 12 overlapping 3-month seasons
- ONI value (°C) for each season

ONI represents the 3-month running mean of sea-surface-temperature anomalies in the Niño 3.4 region.

Source 6: CDC/ATSDR Social Vulnerability Index (SVI)

Source: CDC / Agency for Toxic Substances and Disease Registry (ATSDR)

Dataset: CDC/ATSDR SVI Data Download

Contains

The SVI measures how vulnerable communities may be when facing emergencies or disasters using 16 social factors grouped into four themes:

- **Socioeconomic Status:** poverty, unemployment, income, education
- **Household Characteristics:** age, disability, single-parent households, English-language proficiency
- **Racial & Ethnic Minority Status**
- **Housing Type & Transportation:** multi-unit housing, mobile homes, crowding, lack of vehicles, group quarters

Source 7: FEMA National Shelter System

FEMA provides the shelter information through an ArcGIS Feature Service rather than a simple downloadable CSV on the page reviewed.

Source: FEMA / Mass Care / American Red Cross

Dataset (to be downloaded): FEMA ESF #6 Shelter System

Available fields

- Shelter ID and name
- Address, county, state and FIPS code
- Latitude and longitude
- Evacuation capacity
- Post-impact capacity
- Shelter status

- Current/general/medical-needs population
- Generator availability
- Wheelchair accessibility
- Pet accommodations
- Floodplain and storm-surge indicators